



Work Breakdown Structure (WBS)

1.0 Preparation & System Design

1.1 Project Documentation

1.2 Project Planning

1.2.1 Project Specifications

1.2.2 User Stories

1.2.3 Use Case Diagram

1.2.4 Timeline Development

1.3 System Design

1.3.1 System Workflow

1.3.2 UI/UX Design

1.3.3 Database design

1.3.4 System Architecture

2.0 Backend Development

2.1 Backend Foundation

2.1.1 Express.js Setup

2.1.2 PostgreSQL/MongoDB Connection

2.1.3 API Structure

2.2 User & Authentication

- 2.2.1 Attendee Registration
 - 2.2.2 Admin Login
 - 2.2.3 Role Authorization
- 2.3 RSVP Management
 - 2.3.1 Event Creation
 - 2.3.2 Event Editing & deletion
 - 2.3.3 Event Format Configuration
 - 2.3.4 Attendee & RSVP Management
- 2.4 QR Code Management
 - 2.4.1 Attendee QR Generation
 - 2.4.2 Booth/Checkpoint QR Generation
 - 2.4.3 QR Validation
 - 2.4.4 QR Scan Recording
- 2.5 Attendance & Checkpoints
 - 2.5.1 Entry Checkpoint
 - 2.5.2 Booth Checkpoints
 - 2.5.3 Duplicate Scan Prevention
 - 2.5.4 Attendance History
- 2.6 Dashboard & Reports
 - 2.6.1 Live Attendance Data
 - 2.6.2 Attendance History Data
 - 2.6.3 Report Generation
 - 2.6.4 Report Export
- 3.0 Frontend Development
 - 3.1 Angular Application Foundation
 - 3.1.1 Angular Application Setup

- 3.1.2 Navigation & Layout
 - 3.1.3 Responsive Interface
- 3.2 Attendee Features
 - 3.2.1 Registration & RSVP
 - 3.2.2 QR Code Display
 - 3.2.3 Attendance History
- 3.3 Admin Features
 - 3.3.1 Event Management
 - 3.3.2 Attendee Management
 - 3.3.3 Checkpoint Configuration
- 3.4 QR Scanning
 - 3.4.1 QR Scanner Interface
 - 3.4.2 Entry Scanning
 - 3.4.3 Booth Scanning
- 3.5 Dashboard & Reports
 - 3.5.1 Live Attendance Dashboard
 - 3.5.2 Attendance Reports
 - 3.5.3 Report Export
- 4.0 Quality Assurance & Testing
 - 4.1 Functional Testing
 - 4.1.1 Authentication Testing
 - 4.1.2 RSVP Testing
 - 4.1.3 User Role Testing
 - 4.2 QR & Attendance Testing
 - 4.2.1 QR Generation & Validation
 - 4.2.2 Entry Scanning
 - 4.2.3 Booth Scanning

4.2.4 Duplicate Scan Handling

4.3 Event Format Testing

4.3.1 Single Day Events

4.3.2 Multi-Day Events

4.3.3 Multi-Stop Events

4.4 System Testing

4.4.1 Dashboard Testing

4.4.2 Report Testing

4.4.3 Security Testing

4.4.4 Usability & Responsive Testing